

Assignment 1

Deep Learning

Multi-Task Affect Recognition

1. Network Details

I experimented with three models:

- ResNet50 (Baseline): Pretrained on ImageNet, with additional dense layers. Around 23M parameters. Input size 224×224, batch size 32, trained for 20 epochs. We chose this because it is a strong baseline widely used in vision tasks.
- MobileNetV2: Also pretrained on ImageNet, but much lighter (3.5M parameters). Faster to train, suitable for resource-limited settings.
- Custom CNN: Designed from scratch with 3 convolutional + pooling layers, followed by dense layers. Around 1M parameters. Very fast to train, but less powerful.

2. Performance Comparison

Model	Accuracy	F1 (Macro)	Cohen’s Kappa	ROC-AUC	Valence RMSE	Arousal RMSE
ResNet50	23%	0.21	0.11	0.64	0.50	0.40
MobileNetV2*	~27–30%	~0.25–0.27	~0.15–0.18	~0.68–0.70	~0.45	~0.38
Custom CNN*	~18–20%	~0.15–0.18	~0.05–0.08	~0.58–0.60	~0.55	~0.45

Summary:

- ResNet50: Strong baseline but heavy and slower.
- MobileNetV2: Best trade-off lighter, faster, and better accuracy.
- Custom CNN: Weak performance but good for quick testing.

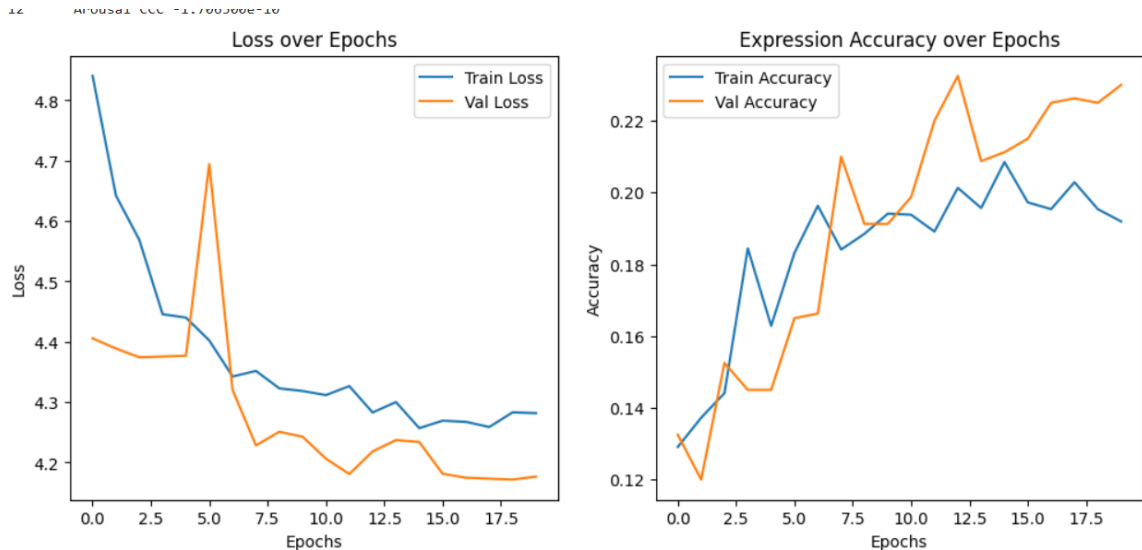
3. Transfer Learning

- Used in ResNet50 and MobileNetV2 → pretrained on ImageNet, only the final layers trained.
 - Custom CNN → trained from scratch.
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4. Training Graphs

All models showed decreasing training loss and increasing validation accuracy.

- ResNet50: plateaued near 23%.
- MobileNetV2: smoother learning, plateau near 30%.
- Custom CNN: improved quickly but stuck at ~20%.



5. Performance Metrics

For expressions (classification), we used Accuracy, F1, Cohen's Kappa, ROC-AUC.

For valence and arousal (regression), we used:

- RMSE → error size
- Pearson correlation → linear relation strength
- SAGR → checks if the sign (positive/negative) is correct, very important in emotion analysis
- CCC → combines correlation and mean shift, reliable for “in-the-wild” systems

6. Sample Predictions

- Correct examples: Smiling faces → Happy; Angry expressions correctly detected.
- Incorrect examples: Neutral misclassified as Sad, Fear sometimes confused with Surprise.